Question 1 – The Doppler Ball

(a)(i)(A) Horizontal component of velocity ✔1 mark

Initial velocity: $v = 11.0\,\mathrm{m/s}$
Angle: $\theta = 36.0^\circ$

$$v_x = v\cos\theta = 11.0\cos(36.0^\circ) = 8.90\,\mathrm{m/s}$$

(a)(i)(B) Vertical component of velocity ✔1 mark

$$v_y = v\sin\theta = 11.0\sin(36.0^\circ) = 6.47\,\mathrm{m/s}$$

(a)(ii) Horizontal distance travelled ✔3 marks

Time of flight: $t = 1.53\,\mathrm{s}$

$$s = v_x \cdot t = 8.90 \times 1.53 = 13.6\,\mathrm{m}$$

(a)(iii) Height of ball above student B ✔4 marks

Time: $t = 0.95\,\mathrm{s}$
Vertical motion from $h = 1.60\,\mathrm{m}$, upwards with $u_y = 6.47\,\mathrm{m/s}$

$$s = ut + \frac{1}{2}at^2 = 6.47 \cdot 0.95 - \frac{1}{2} \cdot 9.8 \cdot (0.95)^2 = 6.15 - 4.42 = 1.73\,\mathrm{m}$$

$$\text{Total height} = 1.60 + 1.73 = 3.33\,\mathrm{m}$$

$$\text{Height above student B's hand (2.10m)} = 1.23\,\mathrm{m}$$

(b)(i) Doppler shift: frequency heard by Student B ✔3 marks

Source frequency: $f = 622\,\mathrm{Hz}$
Source speed: $v_s = 8.60\,\mathrm{m/s}$
Speed of sound: $v = 340\,\mathrm{m/s}$

$$f' = \frac{v}{v - v_s} f = \frac{340}{340 - 8.60} \cdot 622 = 638\,\mathrm{Hz}$$

(b)(ii) How foam protects the circuit board ✔2 marks

> The foam **increases the time** over which the collision occurs, so the **rate of change of momentum** is reduced. This reduces the **impact force** on the circuit board.

🧠 Revision Tips

- **Projectile motion** splits into horizontal and vertical components: treat them independently
- **Vertical motion** uses $s = ut + \frac{1}{2}at^2$ or $v = u + at$
- **Doppler effect** for a moving source: use $\frac{v}{v-v_s}f$ when moving *towards* observer
- Always state physical principles clearly in explanations — especially for 2-mark questions